Qixin Xu

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EDUCATION

Tsinghua University Sept. 2023 – June 2027 (expected)
B.S. in Computer Science Beijing, China

- **GPA:** 3.9/4.0 **Rank:** 8/172
- **Selected A$^+$/A Courses:** Linear Algebra, Probability and Statistics, Introduction to Artificial Intelligence, Introduction to Statistical Learning, Deep Learning, Multi-Modal Machine Learning, Data Mining, Programming and Training, Writing and Communication, Research Paper Writing
- **Research Interests:** Reinforcement Learning, LLM Agents

University of California, Berkeley Jan. 2026 – May 2026
Exchange Student Berkeley, CA, USA

PUBLICATIONS

* Equal contribution † Corresponding author

[1] **Qixin Xu**, Yuxiao Qu, Ian Wu, *et al.*, Aviral Kumar†. **How to Train Memory Writing for Continual Learning**. *In progress*, 2026.

[2] **Qixin Xu**, Fan Nie, Bohan Lyu, James Zou†. **COVE: Co-Evolution through Shared Experience for LLM Agents**. *Under review*, 2026.

[3] **Qixin Xu***, Haozhe Wang*, Che Liu, Fangzhen Lin, Wenhu Chen†. **UniDoc: Towards Unified Thinking in Documents**. *TMLR 2026*.

[4] **Qixin Xu**, Bohan Lyu, Zihan Zhu, Yesai Wu, Zhong Zhang, Haotian Chen, Xin Cong, Xiaojiang Liu, Zhiyuan Liu†, Maosong Sun. **Diversity-aware Training for Test-time Scaling**. *EMNLP 2026*.

[5] Haoxiang Zhang*, **Qixin Xu***, Zhuofeng Li, Lei Zhang, Pengcheng Jiang, Yu Zhang†, Julian McAuley†. **Masking Stale Observations Helps Search Agents–Until It Doesn't: A Regime Map and Its Mechanism**. *EMNLP 2026*.

[6] Haozhe Wang, **Qixin Xu**, Changpeng Wang, Taofeng Xue, Chong Peng, Wenhu Chen†, Fangzhen Lin†. **Bad Seeing or Bad Thinking? Rewarding Perception-Reasoning Synergy for Scalable Vision-Language Reasoning**. *ICML 2026* **(Oral)**.

[7] Bohan Lyu*, Yucheng Yang*, Siqiao Huang*, Jiaru Zhang*, **Qixin Xu***, Xinghan Li*, Xinyang Han*, Yicheng Zhang*, Huaqing Zhang*, *et al.*, Chi Jin†. **MLS-Bench: A Holistic and Rigorous Assessment of AI Systems on Building Better AI**. *NeurIPS 2026* **(Oral)**.

[8] Ganqu Cui*, Lifan Yuan*, Zefan Wang*, Hanbin Wang*, Wendi Li*, Bingxiang He*, Yuchen Fan*, **Qixin Xu***, Tianyu Yu*, *et al.*, Ning Ding†. **Process Reinforcement Through Implicit Rewards**. *TMLR 2026*.

[9] Haozhe Wang, **Qixin Xu**, Che Liu, Junhong Wu, Fangzhen Lin†, Wenhu Chen†. **Emergent Hierarchical Reasoning in LLMs Through Reinforcement Learning**. *ICLR 2026*.

[10] Haozhe Wang, Haoran Que, **Qixin Xu**, Minghao Liu, Wangchunshu Zhou, *et al.*, Ge Zhang†, Fangzhen Lin†. **Reverse-Engineered Reasoning for Open-Ended Generation**. *ICLR 2026*.

[11] Tianyu Yu, Haoye Zhang, Qiming Li, **Qixin Xu**, Yuan Yao†, Da Chen, Xiaoman Lu, Ganqu Cui, Yunkai Dang, Taiwen He, Xiaocheng Feng, Jun Song, Bo Zheng, Zhiyuan Liu†, Tat-Seng Chua, Maosong Sun. **RLAIF-V: Open-Source AI Feedback Leads to Super GPT-4V Trustworthiness**. *CVPR 2025* **(Highlight)**.

[12] Pan Lu*, Yifan Gao*, William G. Peng, Haoxiang Zhang, Kunlun Zhu, Elektra K. Robinson, **Qixin Xu**, Masakazu Kotaka, Harrison G. Zhang, Bingxuan Li, Anthony L. Shiver, Yejin Choi, Kerwyn Casey Huang, Justin L. Sonnenburg†, James Zou†. **Eubiota: Modular Agentic AI for Autonomous Discovery in the Gut Microbiome**. *Submitted to Nature*.

RESEARCH EXPERIENCE

AIRe Lab, Carnegie Mellon University June 2026 – Present
Research Intern, advised by Prof. Aviral Kumar

- **Continual Learning in Sequential Deployment:** Formulated deployment as a stream of related tasks where a frozen-weight agent improves only through a self-written memory; co-trained a single policy with RL to act and write memory, rewarding memory by its downstream benefit. Outperformed all baselines and ablations.

Stanford AI Lab Nov. 2025 – June 2026
Research Intern, advised by Prof. James Zou

- **Co-Evolution of LLM Agents through Shared Experience:** Proposed COVE, where agents on heterogeneous backbones co-evolve via a shared experience memory; showed that cross-model sharing helps while same-model sharing adds little, and introduced mechanisms against negative transfer. First author, under submission.
- **Multi-Agent Systems for Scientific Discovery via Reinforcement Learning:** Developed a multi-agent RL framework for autonomous scientific discovery, achieving state-of-the-art results on domain-specific benchmarks, validated by human evaluation.

TIGER AI Lab, University of Waterloo July 2025 – Nov. 2025
Research Intern, advised by Prof. Wenhu Chen

- **Unified Retrieval and Reasoning over Long Documents:** Proposed UniDoc, a coarse-to-fine framework unifying retrieval and reasoning in one model; showed that direct RL outperforms SFT followed by RL, achieving state-of-the-art results on document reasoning benchmarks. Accepted at *TMLR 2026*.
- **Implicit Mechanisms of Reinforcement Learning:** Identified two-stage RL training dynamics explaining "aha moments" and length scaling; proposed Hierarchy-Aware Credit Assignment, outperforming GRPO across model families. Accepted at *ICLR 2026*.
- **Perception and Reasoning in Multimodal Models:** Characterized the coupling between visual perception and textual reasoning, rewarding perception to improve general visual reasoning. Accepted at *ICML 2026* **(Oral)**.

Tsinghua NLP Lab July 2024 – July 2025
Research Intern, advised by Prof. Zhiyuan Liu

- **Diversity in Large Language Models:** Identified that SFT and RL pipelines reduce generation diversity; developed diversity-aware data curation and noisy embedding injection, improving Pass@N and diversity metrics. Accepted at *EMNLP 2026*.
- **Visual Reasoning in Multimodal LLMs:** Built a tool-use RL framework in veRL for counting and grounding, improving Qwen2.5-VL-7B-Instruct on key benchmarks.
- **PRIME, Implicit Reward Modeling for Reasoning:** Proposed PRIME, updating PRMs online from policy rollouts and outcome labels alone via implicit process rewards. Accepted at *TMLR 2026*.
- **Multimodal Alignment through AI Feedback:** Used feedback from open-source models to reduce VLM hallucination and to guide inference-time scaling. Accepted at *CVPR 2025* **(Highlight)**.

PROJECT EXPERIENCE

MiniCPM-V 2.6, MiniCPM-o 2.6, and MiniCPM-V 4.5 July 2024 – July 2025
Contributor, OpenBMB Team

- Developed an effective pipeline to reduce hallucination in MiniCPM-V 2.6 and MiniCPM-o 2.6
- Designed the pre-training data format enabling streaming capability in MiniCPM-o 2.6
- Technical report accepted at *CVPR 2026*

SKILLS

Languages: English (TOEFL 5.5/6.0; CET-6 651/710), Chinese (native)
Programming: Python, C/C++, SystemVerilog, LaTeX
Tools: PyTorch, DeepSpeed, vLLM, SGLang
RL Frameworks: veRL, Slime, Miles, EasyR1, LLaMA-Factory